NARAYANA IIT ACADEMY INDIA

Sec: OSR.IIT_*CO-SC	GTA-5(P2)	Date: 14-04-23
Time: 3HRS	2016_P2	Max. Marks: 186

KEY SHEET

MATHEMATICS

1	**C**	2	**B**	3	**C**	4	**D**	5	**A**
6	**A**	7	**ABC**	8	**BD**	9	**AC**	10	**A**
11	**ABD**	12	**AC**	13	**CD**	14	**ACD**	15	**A**
16	**A**	17	**B**	18	**A**				

PHYSICS

19	**A**	20	**A**	21	**D**	22	**A**	23	**C**
24	**A**	25	**BD**	26	**BC**	27	**ABCD**	28	**ABC**
29	**AD**	30	**ABCD**	31	**AD**	32	**ACD**	33	**A**
34	**B**	35	**A**	36	**C**				

CHEMISTRY

37	**C**	38	**B**	39	**D**	40	**B**	41	**B**
42	**B**	43	**ABCD**	44	**ABC**	45	**ABCD**	46	**ABCD**
47	**AB**	48	**CD**	49	**BCD**	50	**ABD**	51	**A**
52	**C**	53	**A**	54	**A**				

SOLUTIONS

MATHS

1.

$19x^2+99s+1=0$ roots are s and $\frac{1}{t}$

$\frac{s}{t}=\frac{1}{19}$ $\quad s+\frac{1}{t}=-\frac{99}{19}$ $\quad \frac{st+1}{t}=\frac{99}{19}$

so, $\left|\frac{st+4s+1}{t}\right|=\left|\frac{-99}{19}+\frac{4}{19}\right|=5$

2.

For a fixed positive integer n in range

$\left[n,n+\frac{1}{n}\right), f(x)=n\left|x-n-\frac{1}{2n}\right|=\left|nx-n^2-\frac{1}{2}\right|$

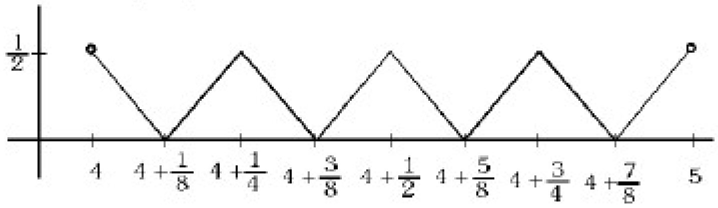
for example when n = 4, $f(x)=\left|4x-16-\frac{1}{2}\right|$

for $\quad 4\le x<4\frac{1}{4}, f(4)=f\left(4\frac{1}{4}\right)=\frac{1}{2}$ and $f\left(4\frac{1}{8}\right)=0$

In general $\quad f(n)=f\left(n+\frac{1}{n}\right)=\frac{1}{2}$ and $f\left(n+\frac{1}{2n}=0\right)$

Thus graph of f on $\left[n,n+\frac{1}{n}\right)$ is like V shape

For $x\in(4,5)$ graph is



3. $\left[\frac{dy}{dx}=\frac{1}{8\sqrt{x}\sqrt{9+\sqrt{x}}}\frac{1}{\sqrt{4+\sqrt{9+\sqrt{x}}}}\right]$

Let $P=\sqrt{4+\sqrt{9+\sqrt{x}}}$

& $\frac{dp}{dx}=\frac{1}{2\sqrt{4+\sqrt{9+\sqrt{x}}}}x\frac{1}{2\sqrt{9+\sqrt{x}}}\frac{1}{2\sqrt{x}}\therefore\frac{dp}{dx}=\frac{dy}{dx}\Rightarrow y=p+\lambda$

& $y(0)=\sqrt{7}\Rightarrow\lambda=0$

$\therefore y(x)=\sqrt{4+\sqrt{9+\sqrt{x}}}\ \ y(256)=3$

4.

$I_{Nr}=\int_0^{\pi/2}\underset{I}{x}\,\underbrace{\cos x.e^{\sin x}}_{II}dx+\int_0^{\pi/2}e^{\sin x}dx$

$$= xe^{\sin x}\Big|_0^{\pi/2} - \int_0^{\pi/2} e^{\sin x}dx + \int_0^{\pi/2} e^{\sin x}dx = \frac{e\pi}{2}$$

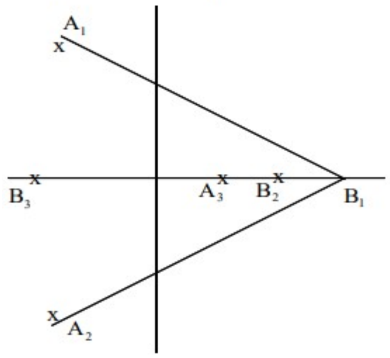
$$\text{Again } I_{Dr} = \int_0^{\pi/2} \underbrace{x}_{I}\,\underbrace{\sin x.e^{\cos x}}_{II}\,dx - \int_0^{\pi/2} e^{\cos x}dx$$

$$= -xe^{\cos x}\Big|_0^{\pi/2} + \int_0^{\pi/2} e^{\cos x}dx - \int_0^{\pi/2} e^{\cos x}dx = -\frac{\pi}{2}$$

$$\therefore \quad \left|\frac{I_{Nr}}{I_{Dr}}\right| = \frac{e\pi.2}{2\pi} = e$$

5. $A = \left\{-1 \pm \sqrt{3}i, 2+0i\right\}$

$B = \left\{8+Oi, \pm 2\sqrt{2}\right\}$. max is A_1B_1 or A_2B_1 $= 2\sqrt{21}$



6. $n\left(\overline{A} \cap \overline{B} \cap \overline{C}\right) = 11! - \{10! + 10! + 10! - (9! + 9! + 9!) + 8!\}$

8. $S = \sum \tan^{-1}\frac{(2r+1)}{2} - \tan^{-1}\left(\frac{2r-1}{2}\right)$

9. $\frac{dy}{dx} - y = -\left(1 - 2x + 3x^2 - 4x^3\right)$

$$ye^{-x} = \int e^{-x}\left(4x^3 - 3x^2 + 2x - 1\right)$$

$$y.e^{-x} = e^{-x}\left(-4x^3 - 9x^2 - 20x - 19\right)$$

10.

$$\begin{vmatrix} \sin(x+A) & \cos(x+A) & a + x\sin A \\ \sin(x+B) & \cos(x+B) & b + x\sin B \\ \sin(x+C) & \cos(x+C) & c + x\sin C \end{vmatrix}$$

$$= \begin{vmatrix} \sin x & \cos x & 0 \\ \cos x & -\sin x & 0 \\ 0 & x & 1 \end{vmatrix} \begin{vmatrix} \cos A & \sin A & a \\ \cos B & \sin B & b \\ \cos C & \sin C & c \end{vmatrix} = -\begin{vmatrix} \cos A & \sin A & a \\ \cos B & \sin B & b \\ \cos C & \sin C & c \end{vmatrix}$$

11. Winning probability of Game A $= \left(\frac{2}{3}\right)^3 + \left(\frac{1}{3}\right)^3 = \frac{27}{81}$

Winning probability of Game B $= \left(\left(\frac{2}{3}\right)^2 + \left(\frac{1}{3}\right)^2\right)^2 = \frac{25}{81}$

12.

$\bar{\alpha} = x(n_1 \times n_2) + y(n_2 \times n_3) + z(n_3 \times n_1)$

Using $\bar{\alpha}.\bar{n}_1 = d_1,\ \bar{\alpha}.\bar{n}_2 = d_2, \bar{\alpha}.\bar{n}_3 = d_3$

$\Rightarrow x = \frac{d_3}{[\bar{n}_1 \bar{n}_2 \bar{n}_3]}, y = \frac{d_1}{[\bar{n}_1 \bar{n}_2 \bar{n}_3]}\ z = \frac{d_2}{[\bar{n}_1 \bar{n}_2 \bar{n}_3]}$

equation of a plane through planes 1 & 2 is

$\bar{r}.(\bar{n}_1 + \bar{n}_2\lambda) = d_1 + d_2\lambda$ which is identical to $\bar{r}.\bar{n}_3 = d_3$

$d_1 + \lambda d_2 = d_3\ \& \bar{n}_1 + \bar{n}_2\lambda = \bar{n}_3$

$\Rightarrow (\bar{n}_1 + \lambda\bar{n}_2) \times \bar{n}_3 = \mu n_3 \times n_3 \Rightarrow -\frac{\bar{n}_1 + \bar{n}_3}{\bar{n}_3 + \bar{n}_3}$

again $\bar{n}_1 + \lambda\bar{n}_2 = \mu\bar{n}_3$

$(\bar{n}_1 + \lambda\bar{n}_2) \times \mu\bar{n}_2 = \mu\bar{n}_3 \times \bar{n}_2$

$\Rightarrow \mu(\bar{n}_3 + \bar{n}_2) = \bar{n}_1 \times \bar{n}_2$

$d_1 + \lambda d_2 = \mu d_3$

$(d_1 + \lambda d_2)(n_2 \times n_3) = d_2(\mu(n_2 \times n_3))$

$d_3(n_2 \times n_3) + d_2(\mu(n_3 \times n_3)) + d_3(n_1 \times n_2) = 0$

13.

$$f(n) = \frac{1}{2(n+1)(n+2)} \sum_{r=0}^{n} {}^{n+2}C_{r+2}(-2)^{r+2} = \frac{(1-2)^{n+2} - 1 + 2(n+2)}{2(n+1)(n+2)}$$

$$= \begin{cases} \frac{1}{n+2} & \text{if } n = \text{odd} \\ \frac{1}{n+1} & \text{if } n = \text{even} \end{cases}$$

14.

$y = mx + \sqrt{4m^2+1}$ passes (0, 2) $\Rightarrow$ $m = \pm\frac{\sqrt{3}}{2}$

$\Rightarrow$ One tangent $\sqrt{3}x + 2y - 4 = 0$

Centre (0, k) $\Rightarrow \frac{|0+2k-4|}{\sqrt{7}} = |k+1| \Rightarrow (k^2 - 4k + 4) = 7(k^2 + 2k + 1)$

$\Rightarrow 3k^2 + 30k - 9 = 0 k^2 + 10k - 3 = 0 \Rightarrow \boxed{k = -5 - 2\sqrt{7}}$

Centre = $(0, -5 - 2\sqrt{7})$

Radius = $|k+1| = 4 + 2\sqrt{7}$

15. Slope of tangent at any point $P(x, y)$ is $\frac{dy}{dx} \propto y \quad \Rightarrow \frac{dy}{dx} = ky \Rightarrow y = e^{k(x-1)}$

$\therefore a(y-1) + (x-1) = 0$ is normal at (1,1) $\Rightarrow k = a$

Hence $f(x) = e^{a(x-1)}$

16. Area bounded = $\int_0^1 \left(\frac{1}{a} - \frac{x}{a} + 1 - e^{a(x-1)}\right) dx \qquad \frac{1}{a}\left(a - \frac{1}{2} + e^{-a}\right)$ sq. unit

17.

$$\text{Locus } C = \left|\frac{\left(x\sqrt{2}+y-1\right)\left(x\sqrt{2}-y+1\right)}{\sqrt{3}}\right| = \lambda^2$$

$2x^2-(y-1)^2=\pm 3\lambda^2$ for intersection with y = 2x + 1

$2x^2-(2x)^2=\pm 3\lambda^2 \quad -2x^2=-3\lambda^2$ (taking –ve sign) $\quad x=\pm\sqrt{\frac{3}{2}}\lambda$

Distance between R and S $=2\left|\sqrt{\frac{3}{2}}\lambda\right|\sec\theta$ ($\tan\theta$ is slope of line) $\quad =\sqrt{6}|\lambda|\sqrt{5}$

So, $\sqrt{30}|\lambda|=\sqrt{360} \quad \lambda^2=12$

18.

Equation of perpendicular bisector $y=-\frac{1}{2}x+1$

For point of intersection $2x^2-\frac{1}{4}x^2=\pm 3\lambda^2$

$x=\pm\sqrt{\frac{12}{7}}\lambda$ (taking +ve sign)

$$\text{Distance} = \left|2.\sqrt{\frac{12}{7}}.2\sqrt{3}.\sec\theta\right| = 2.\sqrt{\frac{12}{7}}.2\sqrt{3}.\sqrt{\frac{5}{2}} = 2\sqrt{3}.\sqrt{\frac{60}{7}}$$

$$\frac{D}{5}=\frac{12\times 60}{7\times 5}=20.57$$

PHYSICS

19. $\mu N \ge \sqrt{(mg\sin\theta)^2+(ma)^2}$

$\Rightarrow \mu mg\cos\theta \ge m\sqrt{a^2+g^2\sin^2\theta}$

$\Rightarrow \mu \ge \sqrt{\tan^2\theta+\left(\frac{a}{g}\right)^2\sec^2\theta}$

By putting a = g/2, and θ = 45°

$\mu \ge \sqrt{\frac{3}{2}}$

20. $v+\frac{v}{4}=\omega R$

21. $\frac{1}{20}=\left(\frac{3}{2}-1\right)\left(\frac{1}{R}-\frac{1}{-R}\right)$

$\Rightarrow R=20cm$

$$\frac{\left(\frac{4}{3}\right)}{v_1}-\frac{1}{(-30)}=\frac{\left(\frac{3}{4}-1\right)}{+20}-\frac{\left(\frac{3}{2}-\frac{4}{3}\right)}{-20}$$

$\Rightarrow v_1=\infty$

After reflection from mirror these rays appear to be passing from focus of mirror

$$\therefore\ \frac{1}{v}-\frac{\left(\frac{4}{3}\right)}{-40}=\frac{\left(\frac{3}{2}-\frac{4}{3}\right)}{+20}-\frac{\left(\frac{3}{2}-1\right)}{-20}$$

$\Rightarrow v=\infty$

22. Find total energy in sphere of radius 2R and then find what stored in cone by using solid angle

23. **Conceptual**

24. $F.\Delta t=3mV_{cm}=3m.\frac{x}{3}\sqrt{\frac{k}{m}}$

25. **Conceptual**

26. **Conceptual**

27. **Conceptual**

28. **Conceptual**

29. By constraint relation, $y+\sqrt{x^2+a}$ = constant

$v_2=v_1\cos\theta$

30. Due to change in frequency saturation current does not change but stopping potential changes.

31. **Conceptual**

32. **Conceptual**

33. After hitting the top plate, the balls will get negatively charged and will now get attracted to the bottom plate which is positively charged. The motion of the balls will be periodic but not SHM.

34. If Q is charge on balls, then $Q\propto V_0$ ……(i)

Also $h=\frac{1}{2}at^2=\frac{1}{2}\left(\frac{QV_0}{mh}\right)t^2$

$\Rightarrow t\propto\frac{1}{V_0}$

Now, $I_{av}\propto\frac{Q}{t}$

$\Rightarrow I_{av}\propto V_0^2$

35. $KE=KE_{com}+(KE\ \ in\ centre\ of\ \ mass\ \ frame)$

36. Draw IAOR

CHEMISTRY

37. It is a concentration cell , $E^0_{cell} = 0$

$$E_{cell} = 0 - \frac{0.06}{1}\log\frac{\left[Ag^+\right]_{LHE}}{\left[Ag^+\right]_{RHE}}$$

$$\left[Ag^+\right]_{RHE} = 0.1M$$

$$0.45 = -\frac{0.06}{1}\log\frac{\left[Ag^+\right]_{LHE}}{(0.1)}$$

$$\frac{0.45}{0.06} = \log\frac{(0.1)}{\left[Ag^+\right]_{LHE}} = 7.5$$

$$\frac{0.1}{\left[Ag^+\right]_{LHE}} = Anti\log(7.5) = 3.16\times10^7$$

$$\left[Ag^+\right]_{LHE} = \frac{0.1}{3.16\times10^7} = 3.16\times10^{-9}M$$

$$\left[Cl^-\right] = 0.2M$$

$$K_{SP}\ of\ AgCl = 3.16\times10^{-9}\times0.2 = 0.63\times10^{-9}M^2$$

38. Negatively charged sol is formed due to the adsorption of SnO_3^{2-} ions.

39. Conceptual

40. $\left[\mathrm{NiCl_4}\right]^{2-}$ is paramagnetic

43. Conceptual

44. Two Cl^- ions have to be removed to create two F centres (electrons occupying the lattice points)

Cl^- can be removed from corners $\frac{1}{8}\times2 = \frac{1}{4}$; $\quad\therefore Na_4Cl_{4-\frac{1}{4}} = Na_4Cl_{3.75}$

Cl^- can be removed from face centres $2\times\frac{1}{2} = 1$; $\quad\therefore Na_4Cl_3$

If one Cl^- is removed from corner and one Cl^- is removed from face centre; then Number of

Cl^- removed $= \frac{1}{8}+\frac{1}{2} = \frac{5}{8} = 0.625$; $\therefore Na_4Cl_{3.375}$

45. Conceptual

50.

51.

$$\begin{array}{cccc} H_3PO_4 \; + & NaOH \rightarrow & NaH_2PO_4 \; + & H_2O \\ 100\times 0.1 & 200\times 0.1 & - & - \\ =10 & =20 & & \\ - & 10 & 10 & \end{array}$$

$$\begin{array}{cccc} NaH_2PO_4 \; + & NaOH \rightarrow & Na_2HPO_4 \; + & H_2O \\ 10 & 10 & - & - \\ - & - & 10 & \end{array}$$

$$pH = \frac{1}{2}\left[pK_2 + pK_3\right] = \frac{7.3 + 10.35}{2} = 8.875$$

52.

$$\underbrace{\begin{array}{cccc} H_3PO_4 \; + & NaOH \rightarrow & NaH_2PO_4 \; + & H_2O \\ 40 & 15 & - & - \\ 25 & - & 15 & \end{array}}_{\text{Buffer solution}}$$

$$\mathrm{pH} = 3.3 + \log\left(\frac{15}{25}\right) = 3.08$$